

CASE REPORT

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Bloodstream infection by *Lactobacillus rhamnosus* in a haematology patient: why metagenomics can make the difference

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Abstract

Background Bloodstream infections (BSIs) pose a persistent threat to hospitalized patients, particularly those who are immunocompromised and susceptible to infections caused by anaerobic or facultative anaerobic bacteria. Alterations in gut microbiota composition can predispose individuals to intestinal domination by one or more pathobionts, increasing the risk of bacterial translocation into the bloodstream and subsequent bacteremia.

Case presentation We report the case of a 20-year-old female with multiple relapsed/refractory Philadelphia-negative B-cell acute lymphoblastic leukemia, initially referred to our hematology center for CAR-T cell therapy. The patient ultimately underwent allogeneic hematopoietic stem cell transplantation, which was complicated by infections, moderate-to-severe graft-versus-host disease, hepatic sinusoidal obstruction syndrome, and transplant-associated thrombotic microangiopathy, all contributing to a fatal outcome. Blood cultures obtained in the final week before the patient succumbed to multi-organ toxicity grew *Lactobacillus rhamnosus*. A fecal sample collected concurrently for intestinal microbiota characterization revealed a marked predominance of *Bacillota* (98.5%), with *Lactocaseibacillus* dominating at 47.9%, followed by *Pediococcus* (18.59%) and *Staphylococcus* (3.5%) at the genus level. We performed genomic comparison between the *L. rhamnosus* isolated from blood cultures and the best-matched strain detected in the intestinal microbiota.

Conclusions We report the isolation of *L. rhamnosus* from blood cultures in a patient post hematopoietic cell transplantation, with genomic similarity to a gut-dominant *L. rhamnosus* strain. This case highlights the potential link between intestinal domination and subsequent bloodstream infection, supporting the value of gut microbiota profiling as an adjunctive tool for monitoring high-risk patients, such as hematopoietic cell transplant recipients.

Keywords *Lactocaseibacillus rhamnosus*, BSI in haematology patients, Intestinal domination, Metagenomics

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Background

Bloodstream infections (BSI) are a constant threat to hospitalized patients due to various risk factors associated with underlying diseases such as cardiovascular disease, cirrhosis, diabetes, and cancer [1], as well as treatment-related factors like exposure to immunosuppressive therapies and the use of indwelling devices [2]. These infections, along with sepsis, represent a leading cause of mortality in patients with hematologic malignancies or solid tumors undergoing intensive cytotoxic chemotherapy, particularly in those with acute leukemia who experience chemotherapy-induced neutropenia [3, 4].

In addition to the risks related to neutropenia post-cytotoxic therapy, hospitalization and inpatient management come with other important risks to the hematologic patient. It is widely known that extensive use of antibiotics confers risk of antibiotic-related adverse events, including antibiotic resistance and toxicity [1] and that long-term hospitalization favors colonization and infection by multidrug resistant strains [2]. What is less well described is the role of the gut microbiota in the pathogenesis of treatment-related morbidity and mortality in haematology inpatients.

The gut microbiota is engaged in a complex relationship with the mucosal epithelium, intestinal epithelial cells, and underlying immune tissues, establishing a dynamic equilibrium that confers resistance to infection and allows the absorption of nutrients [3, 4].

Exogenous factors in the inpatient setting, such as antibiotic use, chemotherapy, and immunosuppressive therapy, can disrupt intestinal microbial equilibrium, resulting in the loss of commensal diversity and in the consequent expansion of potentially pathogenic bacteria (pathobionts [5, 6]. This variation reflects the breakdown of ecological mechanisms that normally constrain pathobionts, including niche competition, colonization resistance, and mutualistic interactions. The depletion of beneficial anaerobes reduces competition for nutrients and space, while also impairing the production of antimicrobial metabolites and immune-modulating signals that maintain mucosal integrity [7]. Finally, immune system impairment further exacerbates this imbalance by altering the metabolic environment, for instance by increasing the availability of oxygen or nitrate, which selectively allows the outgrowth of facultative anaerobes such as *Enterobacteriaceae* [8, 9].

Radiation and chemotherapy, which constitute the conditioning regimen for hematopoietic stem cell transplantation can promote a damage of the gut epithelial stem cells and weaken innate immune defences by affecting host-microbe interactions [3]. Moreover, long-term antibiotic treatment causes a significant change in gut microbiota favouring the overgrowth of pathogens or pathobionts in a process known as intestinal domination

[3, 10, 11], that is commonly recognized when at least 30% of the microbiota by a single predominating bacterial taxon occurs [3]. Recent studies have shown that the increased relative abundance of specific taxa in the gut bacterial communities enhances the risk of bacteraemia in certain settings [3, 12, 13]. Analysis of gut microbiota composition and bacterial genotype may represent a promising tool to assess infection risk in colonized patients [14], although further studies are needed for their use in clinical application [15, 16]. While the effects of intestinal domination of known pathobionts, such as *K. pneumoniae*, have been documented [14]; the potential impact of infections caused by commensals has not been described. *Lactocaseibacillus* spp., formerly known as *Lactobacillus* spp., are a Gram-positive component of human mucosal microflora, including the oral, vaginal and intestinal flora [17–20]. As a result of their beneficial properties, isolates of these species are commonly used as probiotics [17, 21, 22]. When their equilibrium with other microbial components is disrupted, *Lactocaseibacilli* may morph into opportunistic bacteria, giving rise to clinically invasive infections in immunocompromised patients, including abscesses, infectious endocarditis and sepsis [23, 24].

Over the years, a number of BSIs by *Lactobacillus rhamnosus* (*L. rhamnosus*) have been described (Table 1). While the use of probiotics is generally cautious and limited in immunocompromised or critically ill patients, especially in Western healthcare systems, they are occasionally administered as adjuncts, for instance to prevent antibiotic-associated diarrhoea [25, 26]. *L. rhamnosus* has been implicated as an emerging pathogen in patients receiving probiotics, including cardio-surgical patients [27], diabetic patients [28] and in infants [29], raising concerns about their safety [30]. Despite clinical suspicion, no definitive mechanistic explanation for gut translocation or the origin of bloodstream infection has yet been established [31]. In this study, we provide experimental evidence of intestinal domination by *L. rhamnosus*, which was subsequently identified in the bloodstream.

Case presentation

A 20-year-old female was referred to hematology center for CAR-T cell therapy following relapse of Philadelphia-negative B-cell acute lymphoblastic leukemia (ALL). The patient was initially diagnosed with ALL in March 2022 at another institution after presenting with lower limb bruising and marked leukocytosis (WBC 63,710/mm³) with 80% peripheral blood blasts. She received standard multi-agent chemotherapy (LAL1913 protocol), which included an initial steroid and cyclophosphamide pre-phase, followed by induction therapy with vincristine, idarubicin, peg-asparaginase, and dexamethasone, along

Table 1 Literatures review on *L. rhamnosus* blood stream infections, focusing on co-morbidities, numbers of positive blood cultures, sex, catheter presence and typology, use of Next generation sequencing approach for bacterial samples characterization and microbiota profiling

Patient	Disease	N° positive blood cultures	Probiotics	Sex	Bacteria	Catheter	NGS	Reference
Preterm infant	Necrotizing enterocolitis	2	Yes	Female	<i>L. rhamnosus</i> GG (ATCC 53103)	PICC	Yes	(Chiang et al., 2021)
83-year-old	Acute respiratory failure; Haemorrhagic shock due to polytrauma	5	Yes	Male	<i>L. rhamnosus</i>	CVVH	No	(Mikucka et al., 2022)
74-year-old	Acute respiratory failure; Mitral valve; replacement; Tricuspid valve annuloplasty; Diarrhoea	5	Yes	Female	<i>L. rhamnosus</i>	Not reported	No	(Mikucka et al., 2022)
54-year-old	Diabetes	Not reported	Yes	Male	<i>L. rhamnosus</i>	Not reported	No	(Zein et al., 2008)
29-year-old	Hyperthyroidism	2	Unlikely	Male	<i>L. rhamnosus</i>	Venepuncture sites	No	(Dhar et al., 2019)
24-year-old	Replacement bicuspid aortic valve	2	Yes	Female	<i>L. rhamnosus</i> 10330/1 <i>L. rhamnosus</i> 10330/2	Not reported	No	(Kochan et al., 2011)
56-year old	Multi-traumatised	4	Yes	Female	<i>L. rhamnosus</i> GG	CVC	Yes	(Rubin et al., 2022)
64-year-old	Severe active ulcerative colitis	Not reported	Yes	Female	<i>L. rhamnosus</i> GG	Not reported	No	(Meini et al., 2015)

with prophylactic intrathecal methotrexate, cytarabine, and prednisone until April 2022. The patient subsequently underwent seven additional cycles of induction and consolidation therapy until December 2022, achieving measurable residual disease (MRD) negativity. Maintenance therapy began in January 2023 and continued until January 2024, when she experienced her first relapse.

Upon relapse confirmation (85% blasts on bone marrow morphology; CD22 positivity confirmed by flow cytometry), the patient received a single cycle of Inotuzumab as a bridge to CAR-T, once again achieving MRD negativity by flow cytometry. Following leukapheresis for CAR-T manufacturing, a second Inotuzumab cycle was administered, with MRD negativity confirmed via flow cytometry and molecular testing. The patient received CAR-T cell infusion in April 2024, following standard lymphodepleting chemotherapy with fludarabine and cyclophosphamide. Her post-CAR-T course was uneventful, and she remained in molecular remission until July 2024, when MRD re-emerged (10^{-3}).

In response to molecular relapse, she was started on blinatumomab therapy, which was initially complicated by neurotoxicity. Treatment was temporarily halted, and the patient received antiepileptic therapy and high-dose corticosteroids. Blinatumomab was then resumed on an outpatient basis as a bridge to transplant, along with intrathecal chemotherapy, steroids,

and weekly vincristine. This outpatient period was further complicated by SARS-CoV-2 infection, treated with nirmatrelvir.

The patient was admitted for allogeneic hematopoietic stem cell transplantation in early October 2024. A bone marrow aspirate at admission confirmed morphologic persistence of disease, prompting the administration of vincristine and corticosteroids. Due to persistent SARS-CoV-2 positivity on day 9 of admission, she received a course of remdesivir based on infectious disease consultation. Admission microbiological testing for gastrointestinal pathogens (bacteria, viruses, and parasites), fungi (germ tube test), multidrug-resistant organisms, and cytomegalovirus was negative.

Conditioning therapy commenced on day 26 and included total body irradiation (2 Gy twice daily on days -8 to -6), fludarabine (40 mg/m² on days -5 to -3), followed by infusion of peripheral blood stem cells from a 7/8 HLA-mismatched unrelated donor. Graft-versus-host disease (GvHD) prophylaxis included anti-thymocyte globulin (ATG), post-transplant cyclophosphamide (50 mg/kg on days +3 and +5), cyclosporine (from day +5), and mycophenolate mofetil (1000 mg twice daily from day +5 to +30). A summary of the clinical course is presented in Fig. 1a.

Post-transplant complications included pulmonary aspergillosis (treated with isavuconazole and caspofungin) and HHV-6 encephalopathy (treated with foscarnet). *Rhinovirus* was also detected. Febrile neutropenia

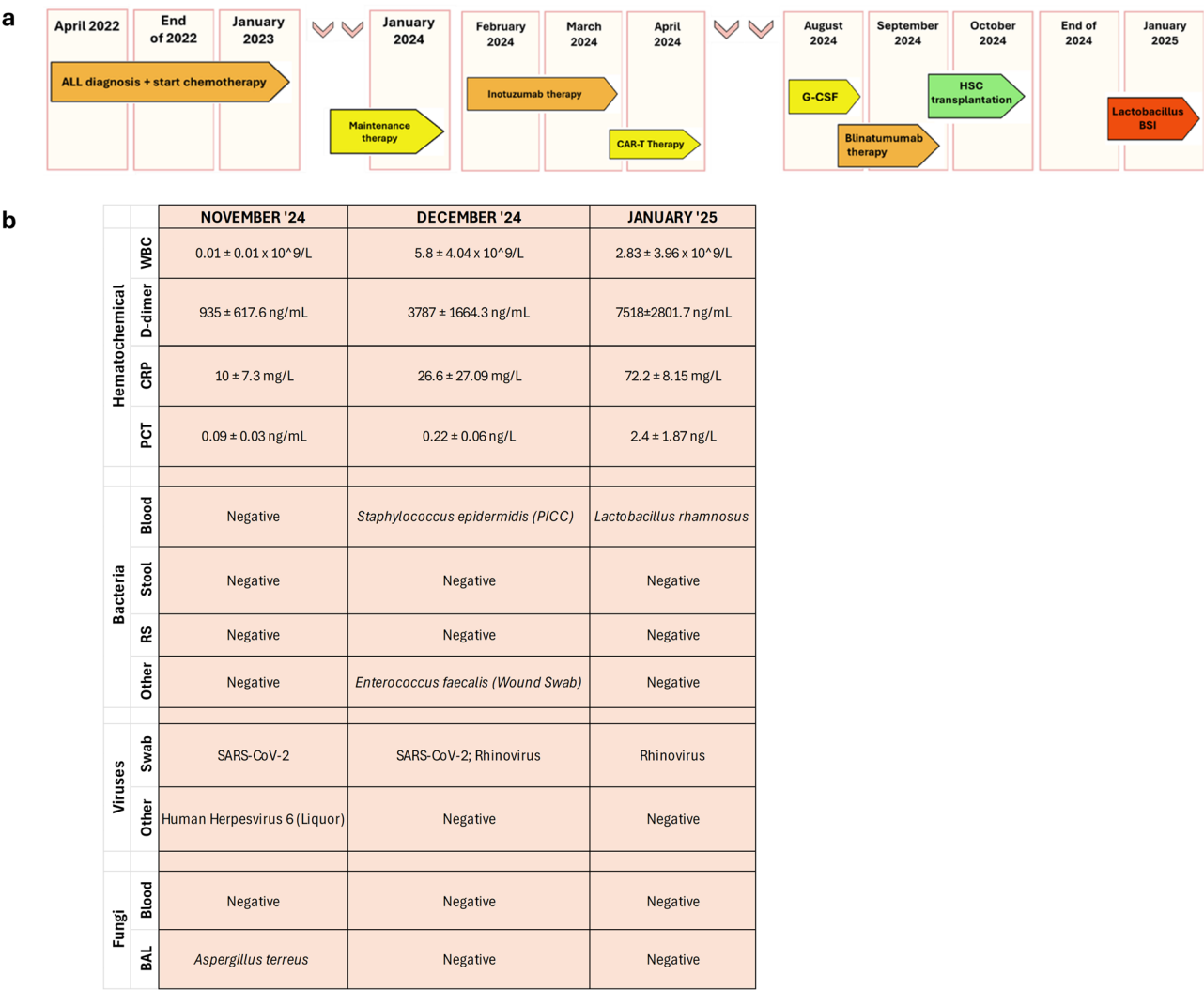


Fig. 1 Schematic timeline reporting information on the clinical history of the patient **(a)**. Patient's blood inflammatory markers and microbiological findings during the period November 2024-January 2025 **(b)**. Panel shows haemato-chemical inflammatory markers sepsis-related: white blood cells (WBC), C-reactive protein (CRP), procalcitonin (PCT). Laboratory microbiological findings regard bacterial, fungal and viral pathogens. Abbreviation: WBC: white blood cells; C-reactive protein (CRP); procalcitonin (PCT); RS: Rectal Swab; BAL: Broncho-Alveolar Lavage

and catheter-related bloodstream infection (BSI) due to *Staphylococcus epidermidis* occurred during conditioning and were treated with meropenem, vancomycin, and liposomal amphotericin B. An *Enterococcus faecalis* infection was treated with cefepime, which was later discontinued due to neurotoxicity. Microbiological findings and inflammatory markers are presented in Fig. 1b.

In January 2025, the patient experienced additional severe transplant-related complications, including acute and chronic GvHD (maximum grade 3 cutaneous and gastrointestinal), for which she received corticosteroids, ruxolitinib, octreotide, and biweekly therapeutic plasma exchange. She also developed sinusoidal obstruction syndrome (treated with defibrotide) and transplant-associated thrombotic microangiopathy (TA-TMA), for which she received fresh frozen plasma infusions.

Despite maximal therapeutic and supportive interventions, the patient died of multiorgan toxicity on day +76 post-transplant. Beginning on day +70 of admission, blood cultures drawn from peripheral veins, peripherally inserted central catheter (PICC), central venous catheter (CVC), and midline catheter were all positive for *Lactobacillus rhamnosus* (*L. rhamnosus*). During this time, inflammatory markers associated with sepsis, including D-dimer and procalcitonin, were markedly elevated (Fig. 1b). On day +70, aerobic and anaerobic blood culture bottles were incubated using the BACT/ALERT® 3D system (bioMérieux, Marcy l'Étoile, France). Three aerobic cultures (one from PICC, two from CVC) became positive after 46 h, revealing Gram-positive rod-shaped bacteria. Identification using the MALDI BioTyper® system (Bruker Daltonics, Bremen, Germany) confirmed *L.*

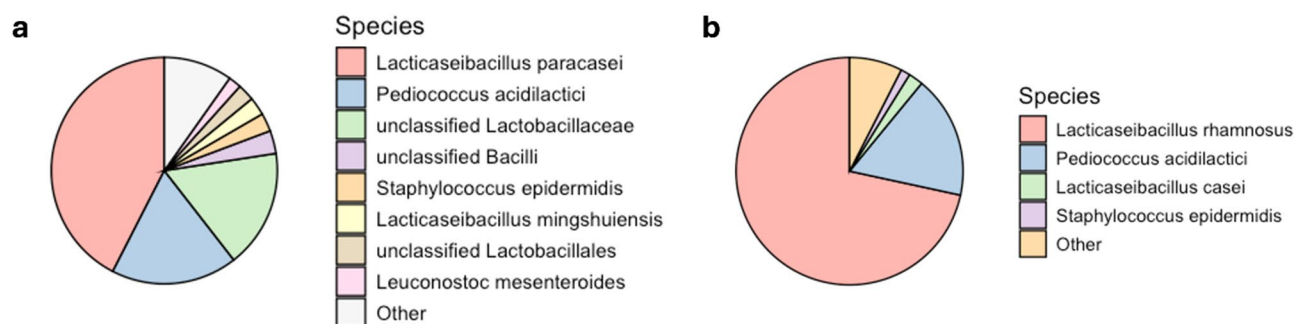


Fig. 2 Gut Microbiota characterization at the moment of *L. rhamnosus* isolation from blood cultures. Panel **a** shows 16 S rRNA gene-targeted intestinal microbiota profiling by using V3-V4-V6 hypervariable regions, while panel **b** shows shotgun metagenomics results

rhamnosus with a score of 2.38 (reliable identification: >1.699). The second and third best matches also identified *L. rhamnosus* (scores: 2.31). Positive cultures were subcultured on Tryptic Soy Agar (TSA) and Columbia agar and incubated at 37 °C with 5% CO₂, confirming the identification.

The genome of the clinical isolate was sequenced (see Additional File 1: Methods). Upon detection of *L. rhamnosus* in blood cultures, a faecal sample was collected and subjected to 16 S rRNA gene-targeted gut microbiota profiling (see Additional File 1: Methods). As shown in Fig. 2a, sequencing of the V3-V4-V6 regions revealed highest relative abundances for *L. paracasei* (42.4%), *Pediococcus acidilactici* (18.7%), and unclassified *Lactobacillaceae* (16.8%). Given that 16 S sequencing cannot reliably differentiate species within the *Lactobacillus casei* group, shotgun metagenomic sequencing was performed. This analysis revealed *L. rhamnosus* as the dominant species (71.6%), followed by *Pediococcus acidilactici* (17.4%) (Fig. 2b). Taxonomically assigned reads corresponding to *L. rhamnosus* were extracted and assembled for genomic comparison with the clinical blood isolate. The two reconstructed genomes differed by only 18 single nucleotide polymorphisms (SNPs), strongly suggesting a clonal relationship. These findings demonstrate gut domination by *L. rhamnosus* and subsequent bloodstream infection, indicating a potential translocation event and highlighting the importance of intestinal microbiota surveillance in high-risk patients.

Discussion and conclusions

Clinical management of immunocompromised patients, in particular hemopoietic cell transplant patients, remains critical, due to their increased susceptibility related to long-term hospitalization and consequent enhanced exposure to multi-drug-resistant bacteria [2, 32] and opportunistic pathogens [1, 12]. In our patient, severe impairment of immune functions may have contributed to *L. rhamnosus* BSI with an unfavourable outcome.

Risk factors in our patient included the acute hematologic malignancy, prior treatment, included B cell-depleting therapy, which directly diminishes host immune defences [1], and extensive use of antibiotics, which promotes bacterial colonization and infection. Together, these factors resulted in dysbiosis of the gut microbiota, favouring the overgrowth of pathobionts [32, 33]. Ultimately, intestinal domination occurred, resulting in occupation of the microbiota by a single predominating bacterial taxon [3], decreasing microbial diversity and increasing the risk of bacterial translocation into the bloodstream [12].

In this study, we demonstrated, for the first time, experimental evidence of intestinal domination by *L. rhamnosus*, which was then rapidly and directly identified from blood specimens [34]. This case demonstrates the role of gut microbiota analysis in high-risk patients, as initial bacterial colonization can give rise to subsequent BSI.

A number of clinical cases on pathobiont-associated BSIs, including *L. rhamnosus*, have been reported, as shown in Table 1 [17, 35]. Although *Lactobacillus*-dominant intestinal microbiota has been reported in some individuals, it remains relatively uncommon in the general population and it may be associated with a specific dietary pattern, probiotic use, or underlying clinical conditions, including polytrauma, severe ulcerative colitis, or hyperthyroidism. This study is the first to show pathobiont-associated BSI mediated by *Lactobacilli* mediated intestinal domination a condition that is not common. Unfortunately, data on intestinal microbiota composition in Italy are limited, even though in European countries, and not only, *Enterococcus* or multi-drug-resistant Gram-negative bacteria intestinal domination is more common [3, 36].

Of note, the use of probiotic supplements and the intake of fermented foods containing *Lactobacillus* (e.g., yogurt, cheese, kefir) were not reported prior to the onset of BSI suggesting that genetic, environmental or metabolic factors may have affected gut microbiome. Interestingly, many of these factors represent important

confounding factors to delineate both healthy and non-healthy gut microbiome status [16].

Our case shows that monitoring the state of eubiosis may be relevant in hematologic patients, who are subject to extensive use of the antibiotics as well as administration of probiotics, including *L. rhamnosus* [21, 37]. Indeed, probiotics can interact with host immune cells in the mucosal epithelium, modulating cytokine secretion and causing systemic effects [38]. *Lactobacilli* can activate macrophages to secrete inflammatory cytokines such as Interleukin-6 (IL-6) and Tumor Necrosis Factor- α [39]. Probiotics can positively modulate B cell activity, as stimulating IL-6 release causes B cell clonal expansion. Some probiotics species may enhance Interleukin-12 production, stimulating development of T-cells and increased NK cell reactivity [38, 40]. Thus, probiotic administration may potentially connect innate and acquired immunity by multiple pathways, though the precise molecular mechanisms remain to be determined. We hypothesize that an overgrowth of these species may induce a significant inflammatory state, which contributes to significant modifications of the host immune response together along with anti-neoplastic treatments [1, 3, 41], and use of other agents, including prophylactic treatment with TMP/SMX [42]. On the other hand, several evidence suggest that COVID-19 may induce significant alterations in the gut microbiota, including a decrease in microbial diversity and an increase in opportunistic pathogens, which can persist even after the acute phase of infection has resolved [43–45]. The loss of the normal equilibrium formed by *Firmicutes* and *Bacteroidota* [46] may predispose to bacterial translocation across the intestinal barrier [47].

Therefore, we acknowledge the potential role of prior COVID-19 infection as a contributing factor to the patient's dysbiosis, in addition to the impact of transplantation-related variables. *Lactobacillus* BSIs are intrinsically resistance to glycopeptides, including teicoplanin and vancomycin [35], even though European Committee on Antimicrobial Susceptibility Testing (EUCAST) breakpoints are not available for *Lactobacilli* [48].

Intriguingly, other probiotic genera, such as *Bifidobacterium*, generally exhibit lower levels of intrinsic resistance to antibiotics and have fewer reported associations with BSIs. For instance, *Bifidobacterium* species show intrinsic resistance to glycopeptides like vancomycin, like *Lactobacillus*, but typically lack acquired resistance genes such as *vanA*, *vanB*, or *vanC* [49–51]. Additionally, novel probiotics like *Akkermansia muciniphila* also displays intrinsic resistance to glycopeptides [51], suggesting that antimicrobial resistance profiles may have significant clinical implications in the management of gut microbiota.

However, potential risk factors specifically for *Lactobacillus* BSIs are difficult to identify: a differential time to positivity between blood cultures obtained from CVC and peripheral vein of about four hours suggests the gut as the source of the BSI, in spite of the potential for *Lactobacillus* ability to form biofilm on catheter lumina [35]. In our study, we were able to demonstrate *L. rhamnosus* intestinal domination, thanks to a complementary analysis of the gut microbiota (metagenomics) together with whole genome sequencing of the clinical isolate.

Lactobacilli may behave as an opportunistic pathogen [23]: several studies have warned against the possible danger of *L. rhamnosus*, which has often been given as a probiotic treatment for intestinal disorders [21]. Their inherent antibiotic resistance profile may selectively promote their growth with respect to other microbial components.

While metagenomic approaches are well established in comparative studies involving two or more groups, their integration into diagnostic workflows remains controversial, largely due to the lack of standardized procedures, normalization strategies, and validated pipelines, particularly in the context of single-sample analyses and biomarkers discovery [16].

We acknowledge the limitations of this study, including the absence of baseline samples which preclude inference of causality regarding the origin of intestinal domination. Nonetheless, our findings join to the growing evidence highlighting a temporal association between intestinal domination and BSI. This case supports the concept that gut microbiota analysis, including relative abundance profiling, could help to identify taxa that may pose a risk for bloodstream infection in high-risk patients, especially those undergoing prolonged hospitalization, broad-spectrum antibiotic therapy, and potential exposure to prebiotic or probiotic interventions, all of which may predispose to the overgrowth of pathobionts.

While sequencing-based approaches (both 16s metabarcoding and shotgun metagenomics) may not yet be practical for a routine clinical use, they offer a comprehensive tool for detecting relevant microbial shifts and for informing the development of faster and accurate targeted assays for early risk stratification and infection surveillance. Notably, many pathobionts are not associated with specific disease syndromes [52] or may only exert a transient impact on the microbiome [16]. As such, they often do not elicit a targeted clinical suspicion and are not routinely included among the etiological agents covered by standard diagnostic panels. In this context, microbiome findings could play a pivotal role in the development of novel diagnostic assays based on quantitative PCR (both real-time and digital PCR). Translation of the metagenomic findings may be essential for the successful integration of these technologies in diagnostic

workflows no less than their ability to outperform current diagnostic methods.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13099-025-00722-3>.

Supplementary Material 1

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Author contributions

This study was designed by CMM, FDM, SS and TDI. Data collection was performed by CMM, JM and FDM. Data analysis and figure preparation were performed by CMM, FDM and GS. CMM and FDM wrote the initial draft. CMM, FDM, JM and SS revised the final manuscript. All authors participated in the discussion and interpretation of the results. All authors have read and agreed to the published version of the manuscript. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

All procedures performed were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Consent for publication

Written informed consent for publication of anonymized data for research and educational purposes was obtained from the patient.

Competing interests

The authors declare no competing interests.

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